

PORT NOLLOTH, South Africa

WMO: 684080

Lat: 29.2500S

Lon: 16.8686E

Elev: 23

StdP: 14.68

Time Zone: 2.00 (E02)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	42.7	44.6	23.3	17.6	69.9	28.2	22.1	68.5	18.9	58.4	17.6	58.3	5.1	100	0.425

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.4	83.3	59.6	78.0	59.2	74.1	60.5	67.3	73.4	66.0	71.5	64.9	69.9	10.0	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.2	93.1	70.2	63.9	89.1	68.8	62.8	85.7	67.8	31.7	72.9	30.8	71.9	30.0	70.1	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.4	16.5	15.1	DB	38.4	98.0	1.3	5.8	37.4	102.2	36.6	105.6	35.9	108.9	34.9	113.1	(4)
(5)			WB	36.7	70.5	1.6	3.1	35.5	72.8	34.6	74.6	33.7	76.4	32.5	78.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	59.8	63.6	63.3	62.6	61.0	58.2	56.4	56.5	55.4	57.0	59.3	60.9	63.2		
	DBStd	5.19	3.18	3.55	3.32	6.04	4.63	5.12	6.66	4.46	3.36	3.60	2.64	2.86		
	HDD50	7	0	0	0	0	0	1	5	1	0	0	0	0		
	HDD65	2088	65	65	91	154	225	269	285	303	246	185	127	73		
	CDD50	3572	422	373	389	331	253	193	207	170	209	289	327	409		
	CDD65	177	21	18	15	35	13	11	22	7	5	9	4	17		
	CDH74	943	14	16	56	278	73	101	150	67	79	91	15	3		
(13)	CDH80	317	0	1	21	114	18	25	40	15	35	43	5	0		
(14)	Wind	WSAvg	6.6	6.7	6.6	6.3	6.3	5.6	6.2	6.7	7.3	7.6	7.6	6.5		
(15) Precipitation	PrecAvg	2.7	0.1	0.1	0.2	0.4	0.3	0.5	0.4	0.4	0.2	0.2	0.1	0.1		
	PrecMax	4.7	1.0	0.4	0.7	1.8	1.8	1.9	1.7	1.2	0.8	1.1	0.6	1.3		
	PrecMin	0.8	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
	PrecStd	1.1	0.2	0.1	0.2	0.5	0.4	0.5	0.4	0.4	0.2	0.2	0.2	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.9	77.3	80.0	92.3	82.7	83.5	84.6	81.8	86.4	87.8	73.3	74.4		
		MCWB	68.5	67.6	66.8	64.9	58.8	58.5	58.8	55.0	59.3	58.2	61.8	66.8		
	2%	DB	72.9	73.3	72.1	81.6	73.9	75.8	78.3	73.8	71.3	72.7	69.2	71.8		
		MCWB	66.3	66.8	64.2	61.7	56.2	55.2	56.2	53.4	54.6	57.2	61.8	65.2		
	5%	DB	70.7	71.0	69.4	74.7	68.5	69.5	72.8	67.4	64.3	66.3	67.5	70.1		
		MCWB	64.8	64.9	63.3	60.2	56.3	52.7	54.2	52.5	56.4	59.2	61.0	64.0		
	10%	DB	69.0	69.1	67.6	67.8	64.4	63.6	67.6	62.2	62.1	64.4	65.9	68.5		
		MCWB	63.8	63.7	62.3	59.5	57.6	54.5	52.5	54.4	55.8	58.4	60.0	62.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.1	68.9	67.5	66.3	63.1	60.7	60.2	58.4	60.1	63.9	64.5	67.3		
		MCDB	74.5	74.7	76.7	86.3	69.9	76.7	77.7	66.1	76.1	76.3	69.0	73.2		
	2%	WB	67.3	66.9	65.5	63.9	60.4	58.5	58.1	57.1	58.2	61.2	62.9	65.7		
		MCDB	71.5	71.9	70.8	75.7	66.1	66.5	70.2	63.7	65.2	67.3	67.7	70.7		
	5%	WB	65.8	65.6	64.1	62.0	59.2	57.4	56.6	56.1	57.2	59.9	61.8	64.7		
		MCDB	69.8	70.0	68.3	69.7	64.2	62.9	65.3	62.0	62.7	65.1	66.4	69.2		
	10%	WB	64.4	64.3	62.9	60.8	58.3	56.4	55.4	55.2	56.3	58.9	60.7	63.6		
		MCDB	68.5	68.2	66.7	66.6	62.8	61.1	62.3	60.3	61.1	63.6	64.9	67.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.4	11.1	10.5	12.6	12.3	14.1	14.8	14.0	12.5	11.4	10.6	10.4		
		MCDBR	11.8	12.4	12.7	24.0	20.0	23.9	24.2	25.1	18.0	14.8	12.6	11.4		
		MCWBR	6.9	7.0	7.0	9.3	9.6	10.2	10.0	11.5	10.1	8.0	7.2	6.6		
	5% WB	MCDBR	10.8	11.4	11.6	15.6	13.4	14.9	16.4	15.0	13.9	12.6	11.0	10.6		
		MCWBR	6.5	6.9	6.8	7.7	8.7	9.8	9.7	10.1	8.8	7.4	6.8	6.3		
(40) Clear-Sky Solar Irradiance	taub		0.353	0.361	0.348	0.326	0.307	0.298	0.306	0.326	0.346	0.355	0.338	0.342		
	taud		2.488	2.486	2.510	2.564	2.582	2.568	2.535	2.473	2.388	2.394	2.493	2.496		
	Ebn at Noon		313	305	299	291	282	277	279	286	294	303	316	318		
	Edh at Noon		37	36	33	28	25	24	26	30	37	39	36	37		
(44) All-Sky Solar Radiation	RadAvg		2600	2369	2021	1574	1221	1075	1163	1420	1817	2231	2564	2709		
	RadStd		110	101	64	67	47	43	49	56	75	71	82	108		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page